SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent
Shaik Roshan-192425258

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr DUMPALA PRASANTH

QUESTIONS: 10 **Total Marks: 50**

Annexure A
Application - Based Questions
Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.
(5 Marks)

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.
(5 Marks)

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

(5 Marks)

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.

(5 Marks)

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance**.** **(5 Marks)**

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

Code of Conduct:

I Shaik Roshan(192425258)*Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SHAIK ROSHAN

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – ANSWER SCRIPT

Course Code	MLA03
Course Name	Reinforcement Learning for Environment and Agent
Course Outcome	CO2: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)
Assessment Tool Weightage	30%
Faculty	Dr Dumpala Prasanth
Annexure	Annexure A - Application-Based Questions
Duration	2 hrs
Total Questions	10
Total Marks	50

This document reproduces the original Application-Based Questions from *CO2_AT2_APPLICATION_BASED_QUESTIONS* followed by complete, well-structured answers for all 10 questions, with RL-framework tables and comparison points for clarity.

Question 1 (5 Marks)

Q1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality.

Answer:

A ride-sharing platform can model driver-passenger matching as an RL problem where a central dispatch agent (or each driver-agent) learns a matching/dispatch policy that minimizes passenger wait time, minimizes driver idle time, and maximizes overall trip completion efficiency.

Component	Description
State Space (S)	Real-time locations of available drivers and waiting passengers, current demand density by zone, traffic conditions, time of day, and driver ratings/availability status.
Action Space (A)	Assign a specific driver to a specific passenger request, reposition an idle driver to a high-demand zone, or hold/queue a request briefly for a better match.
Reward Function (R)	Positive reward for short passenger wait time, short driver idle time, and completed trips; negative reward/penalty for long wait times, driver-passenger mismatch (e.g., far pickup distance), or cancelled rides.

RL Framework Design

- **Agent:** A centralized dispatch controller (or multi-agent system of driver agents) that continuously observes supply-demand state and assigns matches.
- **Environment:** The city road network with dynamically arriving passenger requests and moving drivers.
- **Policy (π):** Learned matching strategy mapping the current driver-passenger state to the assignment action that minimizes cumulative wait time.
- **Learning approach:** Deep RL (e.g., Deep Q-Network or multi-agent RL) is typically used since the state-action space is large and continuously changing.
- **Outcome:** Over many trips, the policy learns demand hotspots and optimal driver repositioning, reducing average wait time and improving service quality/customer satisfaction.

Question 2 (5 Marks)

Q2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations.

Answer:

An e-learning platform models content recommendation as an RL problem: the agent (recommender) selects the next study material for a learner, observes engagement/performance as reward, and updates its policy to personalize future recommendations for better learning outcomes.

Component	Description
State (S)	Learner's current knowledge level, past performance/quiz scores, topics already mastered, and content interaction history.
Action (A)	Recommend a specific study material - a topic the learner is already comfortable with (familiar) or a new/harder topic (exploratory).
Reward (R)	Positive reward for improved quiz scores, course completion, and sustained engagement; negative reward for content skipped, poor performance, or disengagement.

Balancing exploration and exploitation

- **Exploitation:** Recommending content similar to what has worked well for the learner before (familiar topics/format) reinforces mastery and keeps engagement high in the short term.
- **Exploration:** Occasionally recommending new or slightly more challenging material helps discover the learner's true skill boundary and prevents the system from narrowing recommendations too early.
- **Epsilon-greedy or bandit-based strategies** are commonly used - the system mostly exploits proven content but reserves a small probability for exploring new material types.
- **Outcome of balance:** Too much exploitation leads to a stagnant, repetitive learning path; too much exploration overwhelms the learner with difficulty. A well-tuned trade-off personalizes the learning path while still expanding the learner's competence over time.

Question 3 (5 Marks)

Q3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

Answer:

A robotic vacuum cleaner can be modeled as an RL agent that learns to navigate a room, cover the maximum floor area, avoid obstacles/falls, and minimize energy/time spent, purely from sensor feedback as it interacts with its environment.

Component	Description
State (S)	Current position/orientation in the room, sensor readings (proximity to walls/obstacles), battery level, and map of already-cleaned vs. uncleaned areas.
Action (A)	Move forward, turn left/right, reverse, or return to the charging dock.
Reward (R)	Positive reward for cleaning new (previously uncovered) floor area; negative reward for collisions with obstacles, re-cleaning the same spot repeatedly, or getting stuck.

Policy and the role of the value function

- **Policy (π):** The learned mapping from the robot's current sensed state to the best next movement action - essentially the robot's navigation/cleaning strategy.
- **Value function V(s):** Estimates the expected long-term (cumulative) reward of being in a given state, e.g., a position in the room, guiding the robot toward states that lead to more area coverage and away from dead-ends or obstacle-dense zones.
- **How it improves performance over time:** As the robot repeatedly explores the room, it updates its value estimates for different positions/paths. Areas that historically led to efficient coverage get higher value, so the policy increasingly favors those paths.
- **Convergence to efficient behavior:** Over multiple cleaning cycles, the robot's policy converges toward systematic, near-optimal coverage patterns (e.g., boustrophedon/zig-zag paths) instead of random wandering, reducing cleaning time and missed spots.

Question 4 (5 Marks)

Q4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance.

Answer:

In a crypto trading RL system, the agent observes market state and chooses to buy, sell, or hold, with the reward typically tied to profit/loss. However, cryptocurrency markets are extremely volatile and noisy, making reward function design one of the hardest parts of this application.

Component	Description
State (S)	Current price, historical price trends/technical indicators, order book depth, portfolio holdings, and market sentiment signals.
Action (A)	Buy a certain quantity, sell a certain quantity, or hold the current position.
Reward (R)	Typically the realized/unrealized profit or loss from the action, sometimes adjusted for risk (e.g., Sharpe ratio) and transaction fees.

Challenges in designing the reward function

- **Extreme volatility and noise:** Crypto prices can swing sharply on non-fundamental news, making raw profit/loss a noisy, unreliable reward signal.
- **Sparse and delayed rewards:** The true outcome of a trade may only be clear after the position is closed, sometimes much later, complicating credit assignment.
- **Risk-return trade-off:** A reward based purely on profit encourages excessive risk-taking; it must be balanced against volatility/drawdown penalties.
- **Transaction costs and slippage:** If not included in the reward, the agent may learn to overtrade, appearing profitable in backtests but unprofitable after real fees.
- **Non-stationarity:** Market regimes change constantly, so a reward function tuned to one market condition (e.g., bull market) may reward poor decisions in another (bear market).

Impact of poor reward design

- **Reward hacking:** The agent may find unintended ways to maximize reward (e.g., excessive small trades to farm fee rebates) without genuinely improving trading performance.
- **Overexposure to risk:** A pure profit-maximizing reward without risk penalty can produce a policy prone to catastrophic losses during high-volatility events.
- **Poor generalization:** A reward overly tailored to historical data may cause the policy to fail when market conditions shift, leading to real financial losses despite good backtest performance.

Question 5 (5 Marks)

Q5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

Answer:

A smart grid RL system manages electricity distribution by learning to balance supply and demand across the network in real time, deciding how power is routed, stored, or drawn from different sources to maintain reliability while minimizing cost and losses.

Component	Description
State Space (S)	Current electricity demand across zones, generation output (including renewables), battery/storage levels, grid frequency/voltage stability, and electricity market prices.
Action Space (A)	Route power between zones/substations, charge or discharge storage units, curtail non-critical load, or purchase/sell power from/to the external grid.
Reward Mechanism (R)	Positive reward for maintaining supply-demand balance, minimizing transmission losses, and reducing cost; negative reward for blackouts, voltage instability, or excessive reliance on expensive backup generation.

RL Model Summary

- The agent continuously monitors grid state via smart meters/sensors and takes distribution actions every control interval (e.g., every few minutes).
- Reward shaping balances multiple objectives - reliability, cost, and sustainability (favoring renewable usage) - simultaneously.
- Over time, the learned policy adapts to daily/seasonal demand cycles and generation variability, improving grid stability and reducing operating cost more effectively than static rule-based control.

Question 6 (5 Marks)

Q6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes.

Answer:

In a healthcare monitoring system, RL can support personalized dosage adjustment by treating each patient's physiological response as an environment: the agent (decision-support system) observes vital signs, adjusts dosage within safe clinical limits, and learns from patient response over time.

Component	Description
State (S)	Patient's current vital signs (blood pressure, glucose level, heart rate), medical history, current medication level, and recent symptom trends.
Action (A)	Increase, decrease, or maintain the current medication dosage within clinically approved safe bounds.
Reward (R)	Positive reward for vital signs moving toward/staying within the healthy target range and symptom improvement; negative reward for adverse reactions, dosage-related side effects, or vitals moving outside safe limits.

How learning improves outcomes

- **Personalization:** The system learns patient-specific response patterns (since drug response varies by individual), gradually tailoring dosage recommendations rather than applying a one-size-fits-all protocol.
- **Early adaptation:** Continuous monitoring lets the agent detect early signs of an adverse trend and adjust dosage proactively rather than waiting for a scheduled review.
- **Safety constraints:** Because errors are costly, such systems are typically trained with strict safe-RL constraints and clinician oversight, using the reward signal to guide - not fully replace - medical decision-making.
- **Improved long-term outcomes:** As more patient interaction data accumulates, the value function better estimates the long-term benefit of dosage choices, leading to more stable vitals and fewer complications over the treatment course.

Question 7 (5 Marks)

Q7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

Answer:

A drone delivery system uses RL for autonomous navigation: the agent (drone) learns to fly from a depot to delivery destinations while avoiding obstacles (buildings, no-fly zones, weather hazards) and minimizing flight time/energy consumption.

Component	Description
State (S)	Drone's current GPS position/altitude, battery level, wind/weather conditions, obstacle map, and destination coordinates.
Action (A)	Move in a chosen direction/altitude, adjust speed, hover, or return to base for recharging.
Reward (R)	Positive reward for reaching the destination quickly and safely with minimal energy use; negative reward for collisions, no-fly zone violations, or running low on battery mid-flight.

Influence of exploration and exploitation on route optimization

- **Exploration:** The drone occasionally tries new flight paths (rather than only its currently best-known route) to discover shorter or safer paths under changing wind/obstacle conditions.
- **Exploitation:** Once a reliable, efficient route to a frequently visited destination is learned, the drone reuses it to guarantee consistent, fast, and safe deliveries.
- **Balance is critical:** Excessive exploration wastes battery/time testing suboptimal paths; excessive exploitation risks the drone repeatedly using a route that has become unsafe (e.g., new obstacle, changed no-fly zone).
- **Result:** A well-tuned exploration-exploitation strategy (e.g., decaying epsilon-greedy) allows the drone fleet to continuously refine routes as conditions change, converging on efficient, adaptive delivery paths.

Question 8 (5 Marks)

Q8. Design and develop a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.

Answer:

A manufacturing robot performing assembly tasks can use RL to learn efficient movement and assembly sequences, minimizing cycle time and error rate while adapting to variations in parts or workstation layout.

Component	Description
State (S)	Current position of the robotic arm/gripper, position/orientation of components to assemble, assembly stage completed so far, and sensor feedback (force/torque, vision).
Action (A)	Move the arm to a target position, grip/release a component, apply a specific force, or proceed to the next assembly step.
Reward (R)	Positive reward for correctly completed assembly steps and reduced cycle time; negative reward for misalignment, dropped parts, collisions, or defective assembly.

Policy and value function in improving efficiency

- **Policy (π):** The learned control strategy mapping the robot's current sensed state to the next precise motor/gripper action needed to progress the assembly correctly.
- **Value function V(s):** Estimates the expected cumulative reward (successful, fast assembly) from a given intermediate state, helping the robot judge which partial configurations are 'on track' toward efficient completion.
- **Iterative improvement:** As the robot performs many assembly cycles, the value function is refined, allowing the policy to eliminate unnecessary movements, reduce misalignment errors, and shorten the overall assembly cycle time.
- **Outcome:** Over time, the robot converges to a near-optimal, repeatable assembly policy that outperforms manually programmed fixed sequences, especially when part variations or minor line changes occur.

Question 9 (5 Marks)

Q9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance.

Answer:

An online advertising system uses RL to decide which ad to show to which user in which context, learning from click/no-click feedback (and downstream conversions) to maximize long-term advertising effectiveness rather than just immediate clicks.

Component	Description
State (S)	User profile/browsing context, page content, device/time, and recent ad interaction history.
Action (A)	Select which ad (or ad ranking/placement) to display to the user.
Reward (R)	Typically +1 for a click (or a weighted value for conversion), 0 for no click; may be extended to include downstream value like purchase amount.

Reward design challenges

- **Sparse rewards:** Click-through rates are typically low (a small fraction of impressions get clicked), so the agent receives very few positive signals to learn from.
- **Click vs. true value mismatch:** Optimizing purely for clicks can reward 'clickbait' ads that generate clicks but no real conversions or business value, misaligning the reward with the actual business goal.
- **Delayed conversions:** The real value of a click (e.g., a completed purchase) may occur well after the click itself, making it hard to attribute reward correctly in the short term.
- **Bias and fairness:** A pure CTR-maximizing reward can lead the system to repeatedly favor certain ad types/users (popularity bias), reducing diversity and possibly violating fairness or advertiser-balance requirements.

Impact on system performance

- Poorly designed rewards can cause the system to over-optimize for short-term clicks at the expense of long-term user satisfaction and advertiser ROI, ultimately hurting platform revenue and trust.
- Well-designed rewards (blending CTR, conversion value, and diversity/fairness penalties) lead to more sustainable improvements in both user engagement and advertiser outcomes.

Question 10 (5 Marks)

Q10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function.

Answer:

A smart irrigation system uses RL to decide when and how much to water crops, learning from soil and weather sensor feedback to maximize crop yield/health while minimizing water waste and cost.

Component	Description
State (S)	Soil moisture level, weather forecast (rainfall probability, temperature), crop growth stage, and water reservoir/availability level.
Action (A)	Irrigate a specific zone with a chosen water volume/duration, or withhold irrigation for that cycle.
Reward (R)	Positive reward for healthy soil-moisture levels and good crop yield with minimal water use; negative reward for over-watering (waste, root damage, runoff) or under-watering (crop stress).

RL Framework Summary

- The agent (irrigation controller) checks sensor state at regular intervals and decides the irrigation action for each field zone.
- By combining soil-moisture feedback with weather forecasts, the system learns to skip irrigation before predicted rainfall, directly reducing water waste.
- Over multiple growing seasons, the learned policy adapts to crop-specific water needs and local climate patterns, improving both water-use efficiency and yield compared to fixed-schedule irrigation.

Summary: RL Components Across All Applications

Q	Application	Key State	Key Action	Key Reward
1	Ride-sharing matching	Driver/passenger locations, demand	Assign driver, reposition	Wait time, idle time, cancellations
2	E-learning recommendation	Learner performance/history	Recommend familiar/new content	Score improvement, engagement
3	Robotic vacuum cleaner	Position, sensor/obstacle data	Move/turn/dock	Area covered, collision penalty
4	Crypto trading	Price, indicators, holdings	Buy/sell/hold	Profit/loss, risk-adjusted return
5	Smart grid distribution	Demand, generation, storage	Route/store/curtail power	Balance maintained, cost, losses
6	Healthcare dosage	Vitals, history, current dose	Increase/decrease/maintain dose	Vitals in target range, safety
7	Drone delivery navigation	Position, battery, obstacles	Move/hover/return to base	Delivery time, safety, energy
8	Manufacturing assembly	Arm position, part status	Move/grip/release	Cycle time, assembly accuracy
9	Online advertising	User context, history	Select ad to display	Clicks, conversions
10	Smart irrigation	Soil moisture, weather, growth stage	Irrigate/withhold	Yield vs. water used

Note: This answer script has been prepared to help you understand and structure responses to the Application-Based Questions. Please review, personalize the wording, and add your registration details before submission, in line with the assessment's Code of Conduct.